

Fonctions usuelles et primitives

Primitives usuelles

Fonction $f(x)$	Primitive $F(x)$
0	C
λ (constante)	$\lambda x + C$
$x^\alpha, \alpha \neq -1$	$\frac{x^{\alpha+1}}{\alpha+1} + C$
$\frac{1}{x}$	$\ln x + C$
$\frac{1}{x^\alpha}, \alpha \neq 1$	$\frac{x^{1-\alpha}}{1-\alpha} + C$
$\sqrt{x}$	$\frac{2}{3}x^{3/2} + C$
$\frac{1}{\sqrt{x}}$	$2\sqrt{x} + C$
e^x	$e^x + C$
$e^{ax+b}, a \neq 0$	$\frac{1}{a}e^{ax+b} + C$
$\frac{1}{x+a}$	$\ln x+a + C$
$\ln(x)$	$x \ln(x) - x + C$
$\sin(x)$	$-\cos(x) + C$
$\cos(x)$	$\sin(x) + C$
$\tan(x)$	$-\ln \cos(x) + C$
$\frac{1}{1+x^2}$	$\arctan(x) + C$
$\frac{1}{\sqrt{1-x^2}}$	$\arcsin(x) + C$
$\frac{-1}{\sqrt{1-x^2}}$	$\arccos(x) + C$
$\operatorname{sh}(x)$	$\operatorname{ch}(x) + C$
$\operatorname{ch}(x)$	$\operatorname{sh}(x) + C$

Opérations et primitives

Fonction $f(x)$	Primitive $F(x)$
$u'(x)u^n(x), n \neq -1$	$\frac{u^{n+1}(x)}{n+1} + C$
$\frac{u'(x)}{u(x)}, u(x) \neq 0$	$\ln u(x) + C$
$\frac{u'(x)}{u^n(x)}, u(x) \neq 0, n \neq 1$	$\frac{u^{1-n}(x)}{1-n} + C$
$\frac{u'(x)}{\sqrt{u(x)}}, u(x) > 0$	$2\sqrt{u(x)} + C$
$u'(x)e^{u(x)}$	$e^{u(x)} + C$
$u'(x) \cdot u^\alpha(x), u(x) > 0, \alpha \neq -1$	$\frac{u^{\alpha+1}(x)}{\alpha+1} + C$
$\frac{u'(x)}{u^2(x)+1}$	$\arctan(u(x)) + C$